From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 07)** contains 7 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (08 – 11) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Section-A (12 – 16) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 – 07)** contains 7 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

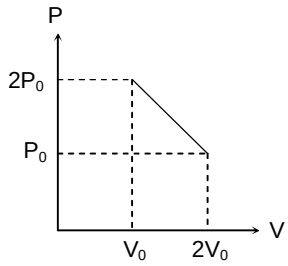
PART – I

SECTION – A

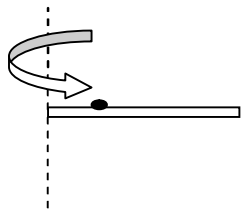
Straight Objective Type

This section contains **7 multiple choice questions** numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

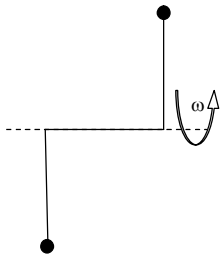
1. A diatomic gas expands from $(V_0, 2P_0)$ to $(2V_0, P_0)$ as shown in the P-V graph. For this process choose the correct statement
(A) Specific heat is positive throughout the process
(B) Specific heat is negative throughout the process
(C) Specific heat is negative from $V = 3V_0/2$ to $V = 7V_0/4$
(D) Specific heat is negative from $V = 5V_0/4$ to $V = 3V_0/2$



2. A uniform straight rod of mass M is free to rotate about a vertical axis through one of its end perpendicular to its length. It is given an angular velocity ω_0 about the vertical axes. And insect of mass M/3 starts moving along the rod away from the axes of rotation with constant speed $L\omega_0$ relative to rod, where L is the length of the rod. By the time insect has reached to the other end, rod has turned through an angle equal to
(A) $\pi/4$ (B) $\pi/2$
(C) $3\pi/4$ (D) π



3. Three massless straight rigid rods, each of length L are rigidly joined to give shape as shown. Two particles each of mass M are fixed at ends as shown. Whole system is free to rotate about a fixed axes passing through the cross section of middle rod as shown in fig. in a gravity free space. What is the net torque acting by the fixed axes on the whole system
(A) Zero (B) $ML^2\omega^2$
(C) $2ML^2\omega^2$ (D) None of these



Space for rough work

4. Six resistor, each of resistance R, are connected at their ends so as form tetrahedral frame. Now two cells of negligible internal resistance, and emf E is connected to the combination. One cell between any two corners and other cell between rest two corners. Thermal power developed will be.

(A) $\frac{E^2}{R}$ (B) $\frac{2E^2}{R}$

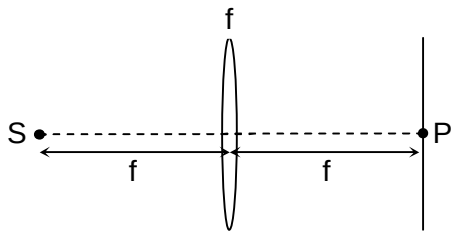
(C) $\frac{4E^2}{R}$ (D) $\frac{8E^2}{R}$

5. Two particles are thrown with same point with same velocity at some time interval. Horizontal component of projection velocity is u. What is Relative angular velocity of the line joining the two particles when they are at same horizontal level.

(A) $\frac{g}{u}$ (B) $\frac{g}{2u}$

(C) $\frac{2g}{u}$ (D) none of these

6. A monochromatic point light source S is placed at a distance 2f from a screen as shown. A convex lens of focal length f is placed in between source and screen as shown. Assume complete transmission of light through lens.



(A) Intensity at point P on the screen before and after placing lens will be same

(B) Intensity at point P on the screen after placing lens will become two times that of before placing lens

(C) Intensity at point P on the screen after placing lens will become four times that of before placing lens

(D) Intensity at point P on the screen before and after placing lens is not comparable

7. A block is suspended by an ideal spring constant K. If the block is pulled down by constant force F and if maximum displacement of block from it's initial position of rest is z, then

(A) z = F/K

(B) z = 2F/K

(C) work done by force F is equal to 2Fz.

(D) increase in potential energy of the spring is $\frac{1}{2}Kz^2$

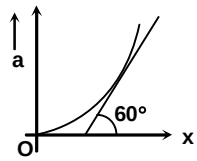
Space for rough work

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

8. Two concentric coil of radii r_1 and r_2 ($r_1 < r_2$) carry currents i_1 and i_2 respectively. If the smaller coil is rotated slightly about one of its diameter it starts oscillating. Then choose the correct statements.
(A) The oscillations are simple harmonic in nature
(B) The frequency of oscillation is proportional to product $i_1 i_2$.
(C) The frequency of oscillation is proportional to square root of r_2
(D) The frequency of oscillation is independent of radius r_1.

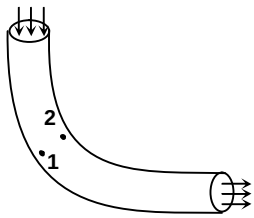
9. A particle starts moving with initial velocity 3 m/s along x-axis from origin. Its acceleration is varying with x in parabolic nature as shown in figure. At $x = \sqrt{3}$ m tangent to the graph makes an angle 60° with positive x-axis as shown in diagram. Then at $x = \sqrt{3}$



(A) $v = \sqrt{(\sqrt{3}+9)}$ m/s (B) $a = 1.5$ m/s^2
(C) $v = \sqrt{12}$ m/s (D) $a = 3$ m/s^2

10. A vertical capillary tube with inside radius 0.25 mm is submerged into water so that the length of its part protruding over the water surface is equal to 25 mm. surface tension of water is 73×10^{-3} N/m and angle of contact is zero degree for glass and water, acceleration due to gravity is 9.8 m/s^2. Then choose correct statement.
(A) R = 0.25 mm (B) h = 59.6 mm
(C) R = 0.60 mm (D) h = 25 mm
where R is radius of meniscus and h is height of water in capillary tube.

11. Ideal fluid flows along a tube of uniform cross section, located in a horizontal plane and bent as shown in figure. The flow is steady, 1 and 2 are two points in the tube. If P_1 and P_2 are pressure at the two points and v_1 and v_2 are the respective velocities, then



(A) $P_1 < P_2$ (B) $P_1 > P_2$
(C) $v_1 < v_2$ (D) $v_1 > v_2$

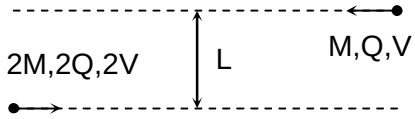
Space for rough work

Comprehension Type

This section contains **2 paragraphs**. Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 12 to 13

Two particles approach each other from very large distance as shown. (consider gravitational force between the particle is negligible)



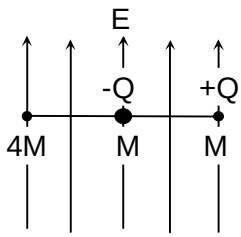
12. If the least separation between the particles is b, what is the speed of M in center of mass frame when particles are closest
(A) LV/b, (B) 2LV/b,
(C) 3LV/b, (D) 4LV/b

13. If $\frac{Q^2}{LMV^2} = 18\pi\varepsilon_0$, find the least separation between the particles
(A) $\left(\frac{3+\sqrt{13}}{5}\right)L$ (B) $\left(\frac{3+\sqrt{13}}{2}\right)L$
(C) $\left(3+\sqrt{13}\right)L$ (D) 2L

Space for rough work

Paragraph for Question Nos. 14 to 16

Two point masses of mass M and 4M are fixed to the end of a massless straight rod. Mass 4M is kept neutral but mass M is given charge +Q. In the middle of the rod another point mass M with charge –Q is fixed. The whole system is kept on a frictionless horizontal surface perpendicular to uniform horizontal electric field as shown.



14. Ratio of speed of 4M and M with charge –Q when both moving
(A) Varies with time (B) 1 : 1
(C) 3 : 1 (D) 1 : 4

15. Ratio of magnitude of acceleration of –Q and +Q
(A) Varies with time (B) 1 : 1
(C) 1 : 2 (D) 1 : 3

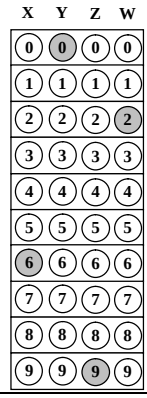
16. Ratio of acceleration of particle at t = 0 to instant when system passes through its stable equilibrium position
(A) Different for different particles (B) 2 : 1
(C) 1 : 2 (D) 1 : 1

Space for rough work

SECTION – C

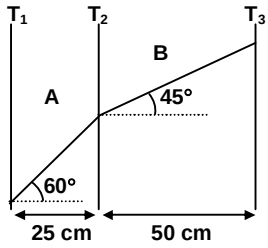
Integer Answer Type

This section contains **7 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.

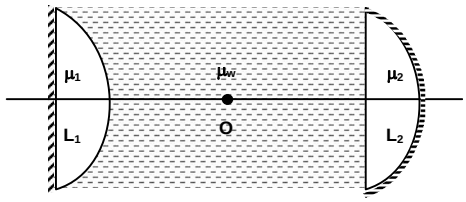


1. A Thread is passing through a hole at the centre of a frictionless table. At the upper end a block of mass 0.5 kg is tied and a block of mass 8 kg is ties at the lower end which freely hanging. The smaller mass is rotated on the table with a constant angular velocity about the axis passing through the hole so as to balance the heavier mass. If the mass of the hanging block is changed from 8 kg to 1 kg. The fractional change in the angular velocity of the smaller mass is $-\frac{K}{4}$, so that it balances the hanging mass again. Find the value of K.

2. Two conductors A and B each of cross section area 5 cm^2 are connected in series. Variation of temperature (in °C) along the length (in cm) is as shown in the figure. If thermal conductivity of A is 200 J/m-sec-°C and the heat current in the conductors is 0.173 J/s. The thermal conductivity of B is $K \times 50\sqrt{3}$ J/m-sec-°C. Find the value of K.



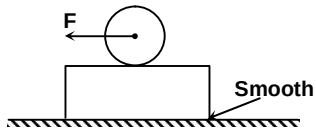
3. A cylindrical tube filled with water (μ_w = 4/3) is closed at its both ends by two silvered plano convex lenses as shown in the figure. Refractive index of lenses L_1 and L_2 are 2.0 and 1.5 while their radii of curvature are 5 cm and 9 cm respectively. A point object is placed some where at a point O on the axis of cylindrical tube.



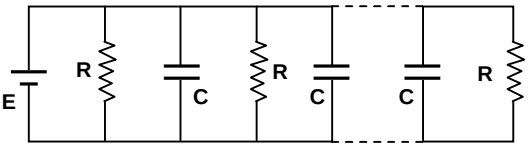
It is found that the object and image coincide each other. Then find the position of object w.r.t. lens L_1 is $K \times 2$ cm. Find the value of K.

Space for rough work

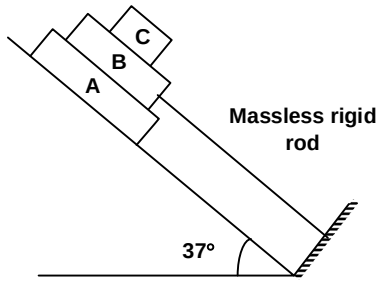
4. A disc of mass m and radius R is placed over a plank of same mass m. There is sufficient friction between the disc and the plank to prevent slipping. A force F is applied at the centre of the disc. Then, the Acceleration of the plank is $\frac{F}{Km}$. Find the value of K.



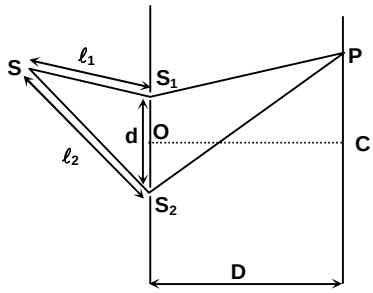
5. n resistances each of resistance R are joined with capacitors of capacity C (each) and a battery of emf E as shown in the figure. In steady state condition, find ratio of charge stored in the first and last capacitor.



6. In the shown figure m_A, m_B and m_C are the masses of the three blocks, the incline is frictionless m_A = 5 kg, m_B = 10 kg, m_C = 2kg. coefficient of friction between A and B is 0.5 and between B and C is 0.1. The frictional force on block A is 6K Newton. Find the value of K. (Take g = 10 m/s^2)



7. In a young double slit experiment. The two slits are illuminated by a monochromatic light source S of wavelength λ = 500 nm. Distance between slits and screen is D = 2m and distance between two slits $S_1S_2 = 10$ μm. ℓ_1 = 20 μm and ℓ_2 = μm, as shown in the figure. If maxima will appear on the screen is 20 K. Find the value of K.



Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **7 multiple choice questions** numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. For the reaction: $3Fe_2O_3(s) \rightleftharpoons 2Fe_3O_4(s) + \frac{1}{2}O_2(g)$,
$\Delta H^o_{298\ K} = 55.5$ kcal and $\Delta G^o_{298\ K} = 46.5$ kcal. What is the equilibrium constant K_p at 298 K for this reaction.
(A) 1.3×10^{-34} (B) 1.3×10^{-35}
(C) 4×10^{-35} (D) 8×10^{-35}

2. If the enthalpy of combustion of benzene, carbon and hydrogen are Q_1, Q_2 and Q_3 respectively, what will be the enthalpy of formation of benzene?
(A) $Q_1 + 6Q_2 + Q_3$ (B) $6Q_2 + Q_1 + 3Q_3$
(C) $6Q_2 - 3Q_3 - Q_1$ (D) $6Q_2 + 3Q_3 - Q_1$

3. $\text{Cyclo pentane} \xrightarrow{Cl_2, h\nu} [X] \xrightarrow[C_2H_5OH]{KOH} [Y] \xrightarrow[H_2O]{O_3/Zn} [Z]$
Here [Z] is:
(A) Pentanal (B) Pentan – 2 – one
(C) Pentane – 3 – one (D) Pentane 1, 5 dial

4. Two liquids A and B have $P^o_A : P^o_B$ = 1:3 at certain temperature. If the mole fraction ratio of $X_A : X_B$ = 1 : 3, the mole fraction of A in vapour in equilibrium with the solution at a given temperature is:
(A) 0.1 (B) 0.2
(C) 0.5 (D) 1.0

5. $A \underset{K_b}{\overset{K_f}{\rightleftharpoons}} B$ is an exothermic reaction, then on increasing temperature,
(A) K_f/K_b, K_f and K_b all increases
(B) K_f, K_b increases and K_f/K_b decreases
(C) K_f decreases, K_b increases, K_f/K_b remains same
(D) K_f, K_b increases and K_f/K_b remains same

Space for rough work

6. Colourless, tetrahedral complexes among the following are:
(A) $K_3[Cu(CN)_4]$ (B) $Na_2[NiCl_4]$
(C) $K[BF_4]$ (D) $Ni(CO)_4$

7. ZnO is white when cold but yellow when hot because
(A) as ZnO is heated its oxidation state changes
(B) due to escape of oxide ions in the form of oxygen, the electrons left in the lattice causes colour to a appear
(C) due to metal deficiency on heating, holes transit, producing colours
(D) due to polarisation of oxide ions

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

8. During the purification of copper by electrolysis:
(A) the anode used is made of copper ore
(B) pure copper is deposited on the cathode
(C) the impurities such as Ag, Au are present in solution as ions
(D) concentration of $CuSO_4$ solution remains constant during dissolution of Cu

9. For the depression of freezing point experiment with non-volative solute, the correct statement(s) is/are:
(A) vapour pressure of pure solvent is more than that of solution
(B) vapour pressure of pure solvent is less than that of solution
(C) only solute molecules solidify at the freezing point
(D) only solvent molecules solidify at the freezing point

10. Select the false statement(s):
(A) Brownian motion and Tyndall effect are shown by true solutions
(B) Sorption process is a combination of adsorption and absorption process
(C) Hardy-Schulze law is related with coagulation of a sol.
(D) Higher is the gold number greater will be the protective power of a lyophilic colloid

11. Which of the following combinations in an aqueous medium will give a blue colour or precipitate?
(A) $Fe^{2+} + [Fe(CN)_6]^{3-}$ (B) $Fe^{3+} + [Fe(CN)_6]^{4-}$
(C) $Co^{2+} + SCN^-$ (D) $Fe^{3+} + SCN^-$

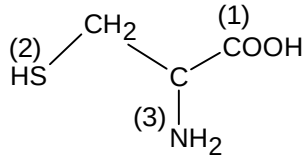
Space for rough work

Comprehension Type

This section contains **2 paragraphs**. Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 12 to 13

Read the following paragraph and answer the questions given below:
The pK_a values for the amino acid cysteine are 1.8, 8.3 and 10.8



12. What is the correct assignment of pK_a values for the functional groups 1, 2 and 3:
(A) 1.8 for 1, 8.3 for 2 and 10.8 for 3 (B) 1.8 for 1, 8.3 for 3 and 10.8 for 2
(C) 8.3 for 1, 1.8 for 3 and 10.8 for 2 (D) 8.3 for 1, 1.8 for 2 and 10.8 for 3

13. What will the pH of cysteine at isoelectric point:
(A) 10.8 (B) 9.55
(C) 5.05 (D) 6.3

Paragraph for Question Nos. 14 to 16

Read the following paragraph and answer the questions given below:
(A) is one of the substance formed when SO_2 and PCl_5 react together. It is a fuming liquid which dissolves in water to give a pungent smelling aqueous solution which turns blue litmus red. 5.95 g of (A) were dissolved in water and the solution was made up to 1 dm^3. Dilute HNO_3 and excess of $AgNO_3$ solution was added to 100 cm^3 of this solution and the precipitate was collected dried & weighed. Dilute HCl followed by $BaCl_2$ solution was added to a second 100 cm^3 portion of solution. No precipitate formed, but on further addition of H_2O_2 solution to the mixture, a white precipitate was obtained. After collecting and drying, the precipitate was weighed.

14. The fuming liquid dissolves in water to give a pungent smelling liquid which turns blue litmus red. The fuming liquid is
(A) SO_2Cl_2 (B) $SOCl_2$
(C) HCl (D) SO_2

15. The weight of precipitate obtained in first case is
(A) 0.2 g (B) 1.435 g
(C) 0.05 g (D) none of these

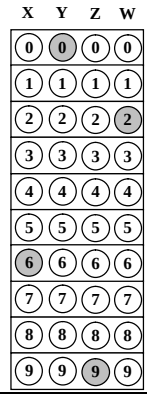
16. The weight of the precipitate obtained in second case is
(A) 0.05 g (B) 1.165 g
(C) 0.583 g (D) 0.595 g

Space for rough work

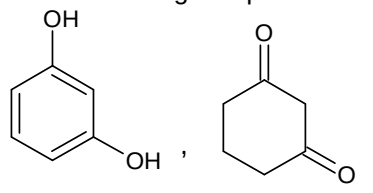
SECTION – C

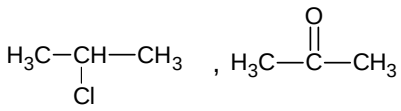
Integer Answer Type

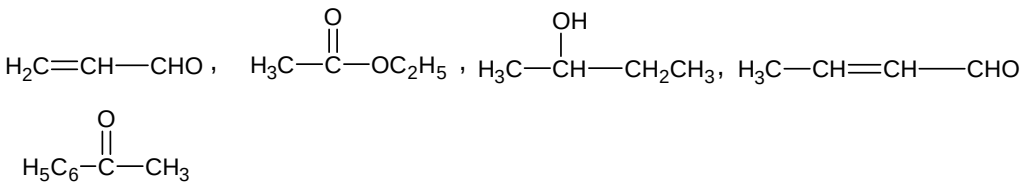
This section contains **7 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. Consider the following compounds:







How many of these compounds would give iodoform test?

2. n-butane on monobromination gives three isomers. The number of product(s) obtained when the major product is heated with KOH in CH_3CH_2OH is/are……..

Space for rough work

3. When a strong current of trivalent gaseous boron is passed through a germanium crystal, its density decreases due to part replacement of Ge by B and due to interstitial vacancies created by missing Ge atoms. When one gm of Ge is taken in the above experiment, the percentage of missing vacancies due to germanium, which are filled by B atoms is found to be 2.376 and boron atoms are found to be 150 ppm by weight. Assuming no volume change in lattice, find the percentage decrease in density of Ge crystal. (At. Wt. of Ge = 72.6, B = 11)

4. What is the minimum pH required to prevent the precipitation of ZnS in a solution that is 0.01 M $ZnCl_2$ and saturated with 0.10 M H_2S? [Given k_{sp} of ZnS = 10^{-21}, For H_2S $K_{a_1} \times K_{a_2} = 10^{-20}$].

5. What will be the simplest stoichiometric coefficient of Cu when it reacts with conc. HNO_3 yielding 3 : 1 ratio of NO_2 and NO.

6. 10 ml of mixture of CO, CH_4 and N_2 exploded with excess of oxygen gave a contraction of 6.5 ml. There was a further contraction of 7 ml when the residual gas was treated with KOH. What is the amount of CH_4 in the original mixture

7. What is ratio of P – O and P = O bonds in $H_4P_2O_7$?

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **7 multiple choice questions** numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. In ΔABC, the ratio $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ is not always equal to (All symbols used have usual meaning in a triangle)
   (A) 2R, where R is circum-radius
   (B) $\frac{abc}{2\Delta}$, where Δ is area of triangle
   (C) $\frac{2}{3}(a^2 + b^2 + c^2)^{1/2}$
   (D) $\frac{(abc)^{2/3}}{(h_1 h_2 h_3)^{1/3}}$ (where h_1, h_2, h_3 are lengths of altitudes)

2. Let α be a real value such that $0 \le \alpha \le \pi$. If $f(x) = \cos x + \cos(x + \alpha) + \cos(x + 2\alpha)$ takes some constant C for any real $x \in R$ then the value of $[C + \alpha]$ is equal to (Where [.] denotes the greatest integer function)
   (A) 0 (B) 1
   (C) –1 (D) 2

3. $\int \frac{\sin 2x + \sin 4x - \sin 6x}{1 + \cos 2x + \cos 4x + \cos 6x} dx$ is equal to
   (A) $\frac{1}{3}\ln|\sec 3x| + \frac{1}{2}\ln|\sec 2x| + \ln|\sec x| + c$ (B) $\frac{1}{3}\ln|\sec 3x| - \frac{1}{2}\ln|\sec 2x| - \ln|\sec x| + c$
   (C) $C - \frac{1}{3}\ln|\sec 3x| + \frac{1}{2}\ln|\sec 2x| + \ln|\sec x|$ (D) $C - \frac{1}{3}\ln|\sec 3x| - \frac{1}{2}\ln|\sec 2x| - \ln|\sec x|$

Space for rough work

4. Let $\beta(x) = \int_0^{x\pi} \sqrt{1-\sin t}\,dt$. The value of $\beta(2) - \beta(1)$ is equal to

(A) $2\sqrt{3}$ (B) $3\sqrt{2}$
(C) 4 (D) 2

5. The centres of the three circles A, B, and C are collinear with the centre of circle B lying between the centres of circle A and C circles A and C both touch externally to circle B and the three circles share a common tangent line. Given that circle A has radius 12 and circle B has radius 42 the radius of circle C, is equal to
(A) 120 (B) 136
(C) 147 (D) 171

6. Let f(x) be a differentiable function in the interval [0, 2] then the value of $\int_0^2 f(x)dx$ is equal to

(A) f(c) where $c \in (0, 2)$ (B) 2f(c) where $c \in (0, 2)$
(C) f′(c) where $c \in (0, 2)$ (D) $\frac{1}{2}f(c)$ where $c \in (0, 2)$

7. Equation of the line in the plane x + 3y – z = 9, which is perpendicular to the line $\vec{r} = \hat{i} + \hat{j} + \hat{k} + \lambda(2\hat{i} + \hat{j} - \hat{k})$ and passing through a point where plane meets the given line, is

(A) $\frac{x-3}{2} = \frac{y-2}{1} = \frac{z}{5}$ (B) $\frac{x-3}{-2} = \frac{y-2}{-1} = \frac{z}{5}$
(C) $\frac{x-3}{-5} = \frac{y-2}{1} = \frac{z}{2}$ (D) $\frac{x-3}{1} = \frac{y-2}{5} = \frac{z}{2}$

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

8. Let $f : R \to [-1, 1]$ be defined as f(x) = cos(sin x) then which of the following is/are correct?
(A) f is periodic with fundamental period 2π (B) range of f = [cos 1, 1]
(C) $\lim_{x \to \pi/2}\left(f\left(\frac{\pi}{2} - x\right) + f\left(\frac{\pi}{2} + x\right)\right) = 2$ (D) f is neither even nor odd function

Space for rough work

9. The sum of the first three numbers in an A.P. is 24 and their product is 384. Then which of the following hold(s) good?
(A) sum of n terms can be $2n(n + 1)$
(B) sum to n terms can be $2n(n + 3)$
(C) sum of n terms can be $14n - 2n^2$
(D) sum of the squares of the three numbers is 224

10. Suppose that f(x) is defined and positive for all real numbers x. If f(x) increases for $x < 0$ and decreases for all $x > 0$, and $g(x) = \frac{1}{f\left(\frac{1}{x}\right)}$ then
(A) g increases for all real x
(B) g increases for $x < 0$
(C) g decreases for $x > 0$
(D) g increases for $x > 0$

11. In ΔABC if $ac = 3$, $bc = 4$ and $\cos(A - B) = \frac{3}{4}$ then
(A) measure of angle A is $\frac{\pi}{2}$
(B) measure of angle B is $\frac{\pi}{2}$
(C) $\cot\frac{C}{2} = \sqrt{7}$
(D) circum–radius of ΔABC is $\frac{2}{7^{1/4}}$

Comprehension Type

This section contains **2 paragraphs**. Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 12 to 13

Read the following write up carefully and answer the following questions:
Let C be the curve $f(x) = (\ln x)^2 + 2 \ln x$ and A(a, f(a)), B(b, f(b)) where $(a < b)$ are the points of tangency of two tangents drawn from origin to the curve C

12. The value of the product ab is equal to
(A) e
(B) $\frac{1}{e}$
(C) e^2
(D) 1

13. Number of values of x satisfying the equations $5x \cdot f'(x) - x \cdot \ln 10 - 10 = 0$ is equal to
(A) 0
(B) 1
(C) 2
(D) 3

Space for rough work

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:

Let f and g be two real valued differentiable function on R satisfying $\int_0^x g(t)dt = 3x + \int_x^0 \cos^2 t \cdot g(t)dt$ and

$$f(x) = \lim_{\alpha \to 0} \frac{1}{\alpha^4} \int_0^{\alpha} \frac{(e^{x+t} - e^x)\ln^2(1+t)}{2t^3+3} dt$$

14. f(ln 2) is greater than

(A) 0 (B) $\frac{1}{6}$

(C) $\frac{3}{20}$ (D) $\frac{4}{25}$

15. Range of g(x) is equal to

(A) $\left[\frac{3}{2}, 3\right]$ (B) $\left[\frac{1}{2}, 3\right]$

(C) $\left[\frac{3}{2}, \infty\right]$ (D) [–3, 3]

16. The value of definite integral $\int_0^{\pi/2} g(x)dx$ has in the interval

(A) $\left(\frac{2\pi}{3}, \frac{4\pi}{5}\right)$ (B) $\left(\frac{\pi}{2}, \frac{5\pi}{6}\right)$

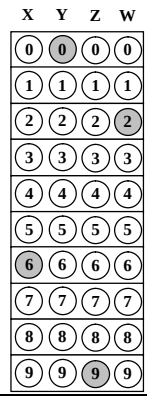
(C) $\left(\frac{3\pi}{4}, \frac{6\pi}{5}\right)$ (D) $\left(\pi, \frac{3\pi}{2}\right)$

Space for rough work

SECTION – C

Integer Answer Type

This section contains **7 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. Let $\sum_{=1}^{\infty}\cot^{-1}\left(\frac{k^2}{8}\right) = -\frac{\pi}{\lambda}$, $\lambda \in I$ then the value of λ is equal to _____

2. Given that for a, b, c, d $\in$ R, if a sec(200º) – c tan(200º) = d and b sec(200º) + d tan(200º) = c then $\frac{a^2+b^2+c^2+d^2}{ac-bd} = p\ \text{cosec}(200^o)$ find the value of p _____

3. Given a triangle ABC with AB = 2 and AC = 1 internal bisector of $\angle$BAC intersect BC at D. If AD = BD and Δ is the area of triangle ABC then the value of $12\Delta^2$ is _____

4. A diagonal of rhombus ABCD is a member of both the family of lines $(x + y – 1) + \lambda_1(2x + 3y – 2) = 0$ and $(x – y + 2) + \lambda_2(2x – 3y + 5)$ where $\lambda_1, \lambda_2 \in R$ and one of the vertex of the rhombus is (3, 2). If area of the rhombus is $12\sqrt{5}$ square units then find the length of semi longer diagonal of the rhombus _____

5. Let f(x) be a function defined on R such that $f'(x) = f'(3 – x)\ \forall\ x \in [0, 3]$ with f(0) = –32 and f(3) = 46, then find the value of $\frac{1}{3}\int_0^3 f(x)dx$ _____

6. Consider an isosceles triangle with base a, vertical angle 20º and lateral side each being b. Then the value of $\frac{a^3+b^3}{ab^2}$ _____

7. Find the smallest integral value of P for which the function f(x) = 6Px – P sin 4x – 5x – sin 3x is monotonic increasing and has no critical point on R _____

Space for rough work